

BIS Sahayak (Bureau of Indian Standards Assistant)

Simple Project Guide & Complete Context • Smart India Hackathon 2026 (SIH26107)

Problem Statement ID: SIH26107	Live Website: https://sih2026-bis-assistant.vercel.app
Theme: Smart Governance & Consumer Safety	GitHub: github.com/PhoenixOP231/BIS_Sahayak_SIH

1. The Real-Life Problem: Why Did We Build This?

Imagine this everyday situation in India:

A Mother Buying a Pressure Cooker or Toys:

- She goes to the market and sees a pressure cooker with a shiny ISI mark stamp.
- *Is it authentic or a fake, dangerous duplicate?*
- Sub-standard pressure cookers can explode, fake electrical wires can catch fire, and cheap uncertified toys can contain toxic heavy metals.
- Everyday citizens have no easy way to know what is safe and what is counterfeit.

A Small Factory Owner (MSME) Making Steel or Cables:

- The Government issues strict **Quality Control Orders (QCOs)** stating that certain products must meet certified Bureau of Indian Standards (BIS) specifications.
- To find the exact rules, the factory owner has to download 100-page, dense government gazette PDF documents full of complex engineering jargon.
- Hiring expensive regulatory consultants costs a lot of money and delays manufacturing.

The Core Problem: Indian Standards are safety-critical and legally mandatory, but they are locked in complicated legal PDFs that regular citizens cannot understand and small businesses struggle to search.

2. What is BIS Sahayak? (The Solution)

BIS Sahayak (Bureau of Indian Standards Assistant) is like a smart, 24/7 digital AI companion for product safety and Indian Standards. It reads and understands thousands of pages of official **Bureau of Indian Standards (IS)** documents and answers any question instantly in plain, easy-to-understand language.

It has **two distinct modes (Dual-Persona)** depending on who is using it:

Consumer Persona (For Everyday Citizens)	Industry / MSME Persona (For Factory Owners & Engineers)
• Plain everyday language explanations • Spotting fake ISI marks & counterfeit warnings • What to check before buying (cookers, water, helmets, toys) • Step-by-step guidance on filing complaints via BIS Care App	• Exact numerical test limits (Proof stress MPa, burst bar, °C) • Mandatory QCO gazette notification dates & compliance rules • Routine vs. Type testing laboratory protocols • Harmonized System (HS) trade classification codes

3. End-to-End User Journey: How People Use It

Scenario A: Everyday Citizen (Consumer Mode)

Goal: Ramesh wants to buy a new pressure cooker and wants to make sure it will not explode.

1. **Open Website:** Ramesh opens <https://sih2026-bis-assistant.vercel.app> on his smartphone.
2. **Select Mode & Language:** He taps **Consumer Mode** and switches language to **English**.
3. **Ask the Question:** He types: **'Is my pressure cooker safe? What should I check?'** (Is my pressure cooker safe? What should I check?)

- **4. Instant Plain Answer:** The AI responds: 'Domestic pressure cookers must be certified under IS 2347:2017. Look for the ISI mark with a 7 or 8-digit CM/L license number printed beneath it.' It gives him a 3-point checklist: (1) Check metallic safety valve, (2) Inspect gasket release system, (3) Never buy without a CM/L number.
- **5. Listen to Audio (🔊):** Ramesh clicks the speaker button and the assistant reads the response out loud in clear Hindi.
- **6. Verify ISI Mark (/verify):** Ramesh flips the cooker box, enters license number **8400123** on the Verify ISI Mark page, and gets instant confirmation of how to check the manufacturer on the **BIS Care App**.

Scenario B: Small Business Owner / Manufacturer (Industry Mode)

Goal: Priya runs a small steel rolling mill in Raipur and wants to manufacture Fe 500D TMT rebars for construction.

- **1. Select Industry Mode:** Priya visits the portal and switches to **Industry / MSME Mode**.
- **2. Ask Technical Query:** She asks: 'What are the mandatory tensile and elongation limits for Fe 500D under IS 1786?'
- **3. Exact Engineering Limits:** The AI immediately outputs the exact mechanical table: Minimum 0.2% Proof Stress / Yield Stress = **500.0 N/mm² (MPa)**, Minimum Tensile Strength (UTS) = **565.0 N/mm²**, Minimum Percentage Elongation = **16.0%**, Mandatory QCO Status = Ministry of Steel gazette order mandates Scheme-I certification.
- **4. Interactive Clause Inspection:** Priya clicks badge **[IS 1786:2008]** right inside the chat. A slide-out drawer appears displaying Clause 8.1 (Chemical composition, Sulphur & Phosphorus limits) and Clause 9.2 (Bend and Re-bend testing parameters) without reloading the page.

4. What Makes It Special? (Key Features)

Feature	What It Does (In Simple Terms)
Zero-Hallucination AI (RAG)	Unlike regular ChatGPT which can guess or make up numbers, BIS Sahayak only answers using authentic, pre-loaded Indian Standards documents. It never invents fake rules.
4-Step ISI License Verifier	A dedicated tool (/verify) where anyone can enter a 7 or 8-digit CM/L number and learn how to confirm the factory's license on the official BIS Care mobile app.
Interactive Standards Directory	A digital catalog (/standards) where you can search through 21+ Indian Standards (helmets, drinking water, toys, LPG cylinders, footwear, electrical cables, cement) with filter tags.
100% Bilingual (English + Hindi)	Complete UI and AI responses in natural Hindi (Devanagari) and English.
Speech Synthesis (Voice Assistant)	Built-in browser audio reading that speaks out answers in natural voices.
100% Uptime & Fast	Built on Next.js 15 and serverless cloud. Even if the external database is offline, a built-in memory engine keeps the app running.

5. How It Works Behind the Scenes (Simplified)

Think of BIS Sahayak like an expert librarian:

Step 1: SEARCH (<15ms) When a user asks a question in English or Hindi, the assistant searches through indexed official BIS Standards (using 768-dim vector math) to locate the exact standard and clause.	Step 2: REASON (Gemini 2.5) Google Gemini 2.5 Flash reads the exact standard clauses and formats the answer for either a Consumer (simple safety tips) or an Engineer (technical tables).	Step 3: PRESENT (Interactive UI) The web app delivers a 2-sentence summary, expandable technical tables, clickable citation badges, and voice playback audio.
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6. Summary: Why This Matters for Smart India Hackathon 2026

- **For Indian Citizens:** Protects families from fake, hazardous goods (fire hazards, exploding cookers, contaminated water, toxic toys).
- **For Indian MSMEs:** Boosts 'Make in India' by making quality compliance fast, easy, and free of expensive consultancy fees.
- **For the Government (BIS):** Increases public awareness of ISI marks and drives higher downloads and usage of the official **BIS Care Mobile App**.